

Big Data Analytics in Smart Manufacturing

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TABLE OF CONTENTS

01

Background

02

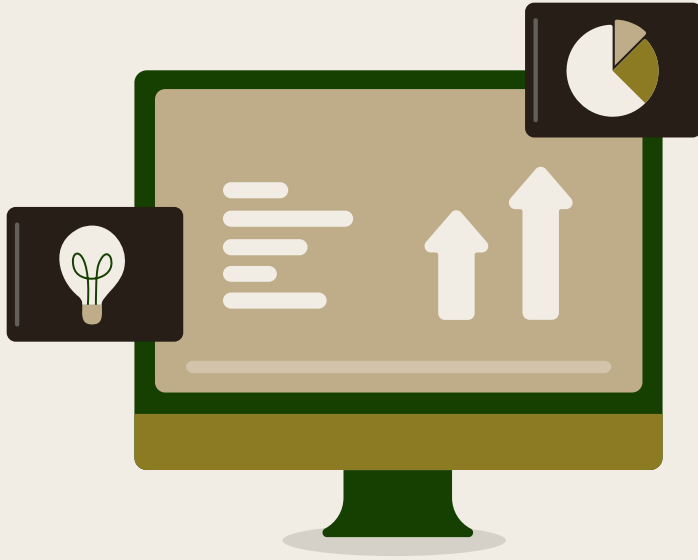
3 Industrial Case
Analytics

03

Strategy &
Journey

04

Recommendation



01 Background

Three Questions Before We Start



01

QUESTION 01

What is Big Data Analytics in modern manufacturing?

02

QUESTION 02

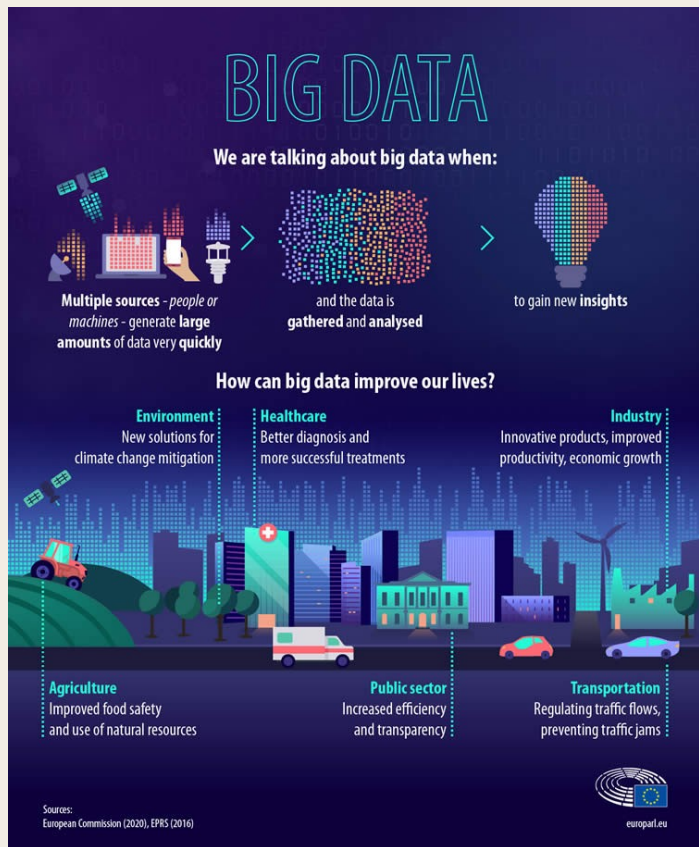
How do enterprises actually deploy and use it?

03

QUESTION 03

What real value does it create on the shop floor?

Background & Architecture



01

Telemetry Ingestion

Ingesting high-velocity telemetry across machines to break operational silos and replace static spreadsheets.

02

Real-Time Insights

Transforming raw factory noise into real-time insights, shifting operations from reactive firefighting to predictive automation.

03

OEE & Value Creation

Translating shop-floor data into direct ROI, maximizing Overall Equipment Effectiveness (OEE) and eliminating unplanned downtime.

The Smart Manufacturing Data Ecosystem

The Industrial Data Pipeline:



Connects shop-floor hardware (Sensors & PLCs) through edge gateways to central enterprise systems (MES & Cloud).



Breaks operational silos by streaming real-time machine telemetry into unified data platforms.

Levels of Analytical Maturity:



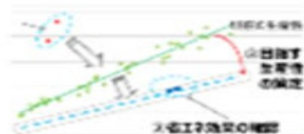
Descriptive (What happened?): Tracking historical machine status and post-shift scrap counts.

Diagnostic (Why did it happen?): Pinpointing root causes of line slowdowns and component overheating.



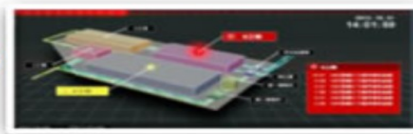
Predictive (What will happen?): Forecasting machine failures hours in advance to eliminate unplanned downtime.

Data analysis and usage



IoT data analysis, machine learning, prediction analysis

Factory visualization Intelligent Dashboard



Integrated data visualization

After-sales Service Application

API Interface



Big data analysis platform
Data integration and accumulation



API Interface

Logistics /Sales Application



Private M2M Cloud
Accumulate performance data for physical equipment / IT systems



SCADA/PLC

Figure: Overview of smart manufacturing project

Object



Product Number Progress

Products Parts



Equipment



Sensor Data

Operations Abnormalities



Process



Performance Results

Processing Quality



People



GPS Screen

Location Activities



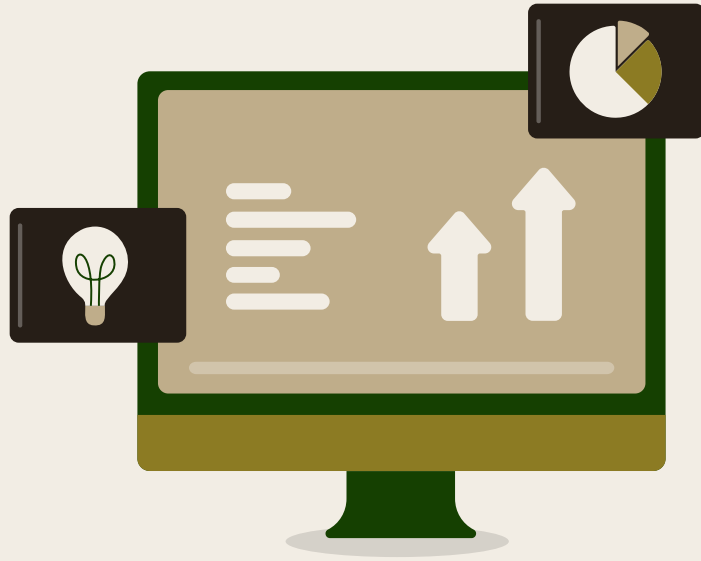
Environment



Usage volume
Sensor measurement

Electricity Temperature





02 3 Case Analysis

CASE STUDY 1

TSMC: In-Line Defect Detection & Yield Defense

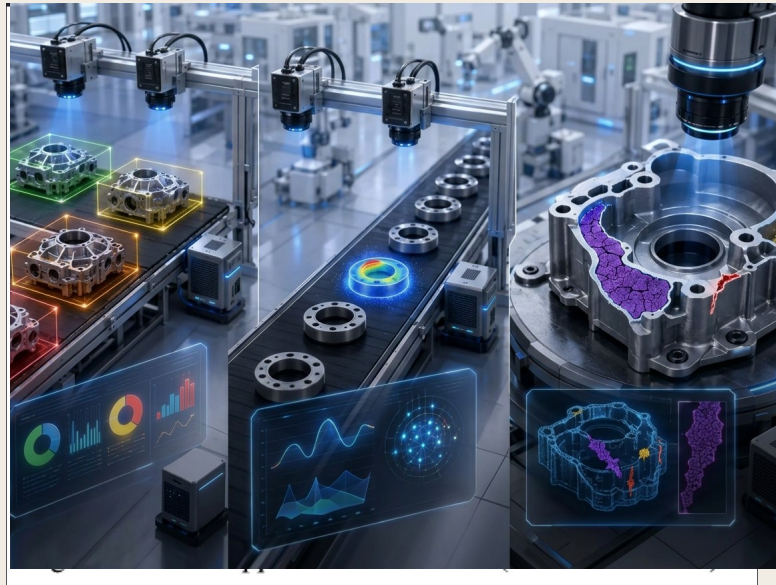


The Operational Challenge:

Advanced packaging and nanometer chips require thousands of sequential steps. A single undetected particle can ruin an entire wafer batch, resulting in catastrophic yield losses.

Measurable Business Value:

Replaces manual post-production inspection with instant prevention, driving wafer yields to industry peaks and saving hundreds of millions of dollars in scrap material.



The Big Data Implementation (How TSMC Uses It):

- 1. Automated Defect Classification (ADC):** Uses edge computing embedded inside machines ("Eagle's View") to analyze millions of images daily and stop faulty tools in real time.
- 2. Product Resume & Traceability:** Encodes unique 2D barcodes on each die to correlate material batches, machine logs, and defect codes for root-cause analysis.

CASE STUDY 2

GE Aerospace: Predictive Maintenance

Operational Challenge

Unplanned engine faults disrupt aircraft availability and maintenance schedules.

Data & Analysis

GE analyzes full-flight data to detect unusual patterns and early component deterioration.

Maintenance Action

Diagnostic engineers review alerts with airlines and plan corrective maintenance.



114 MILLION

flight records per year in Collaborative Insight

GE Aerospace: Repair & Supply Chain Value



5-7 DAYS

Shorter engine-shop turnaround with GE's AI Material Assistant

Parts forecasts months ahead

The tool anticipates parts needed during engine shop visits.

Supply chain implication

Earlier signals supports parts and repair-capacity planning.

CASE STUDY 3

BMW iFactory: Artificial Intelligence in Production

Behind the **iFactory** Technology:

- BMW plants use a cloud-based production IT architecture → **SAP S/4HANA**
 - Product, process, quality, financial, and cost data are linked across the production chain

Data Analytics:

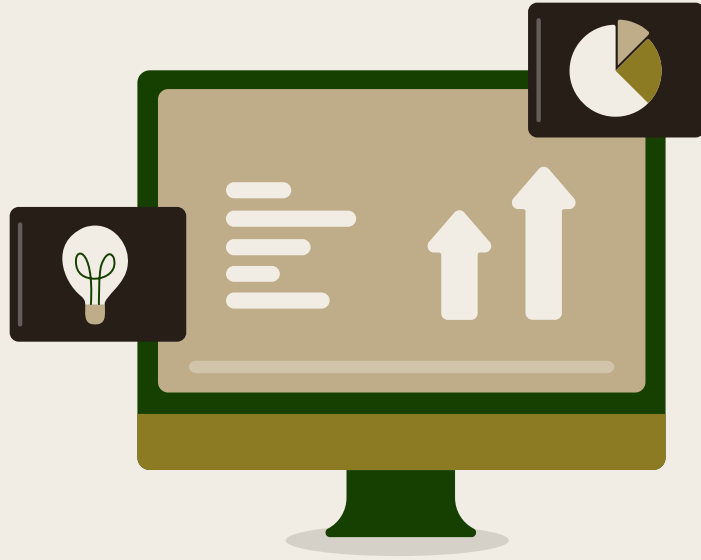
- **AIQX software:** AI-powered quality platform that analyzes real-time sensor and image data to detect production defects and improve quality

Business Value:

- Data creates greater transparency and consistency across plants
- BMW can standardize processes and roll new technology across its global production network faster



**Sensors & Images → AI / Data Analytics → SAP S/4HANA + Cloud
→ Production & Logistics Decisions**



03

Strategy & Journey

Journey and Innovation in Data Analytics for Manufacturing

Past

Collect Data

- Data stored in separate systems and databases
- Sensors and machines generate large volumes of production data

Goal: Capture data + improve visibility

Present

Connect Systems

- Cloud platforms connect production, logistics, quality, and operational data
- Predictive analytics and real time data analysis

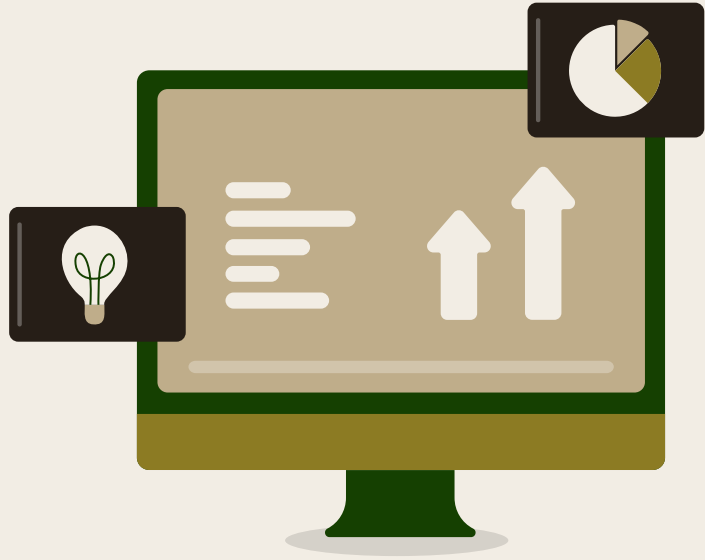
Goal: Consolidate data in one space to anticipate problems

Future

Act Autonomously

- AI-powered systems will coordinate machines, robots, logistics, and production planning
- Humanoid robots perform complex assembly tasks

Goal: Minimal human intervention, AI senses and makes decisions



04

Recommendation

Strategic Recommendations

01

1. A measurable business goal

Choose a costly problem in quality, maintenance or energy. Set a baseline, target and budget.

02

2. Reliable, connected data

Connect machine and production records. Standardize IDs and units, check data quality and secure access.

03

3. People and decisions

Train employees to interpret results. Assign decision owners and require human review for critical actions.

04

4. Evidence before expansion

Measure results against the baseline and total cost. Expand when the pilot meets its agreed targets.



Conclusion

- Manufacturing data supports better decisions about quality, reliability and efficiency.
- Analytics helps teams anticipate problems and improve operations across the product lifecycle.
- Our recommendation: start with a measurable pilot and expand when the benefits justify the cost.

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THANKS!

**Do you have any questions
about Big Data Analytics in
Smart Manufacturing?**

Team 1